1. Instrumental Variables

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i$$

- $X_i = \begin{cases} 0 & \text{if mother } i \text{ is } \underline{\text{not}} \text{ smoking} \\ 1 & \text{if mother } i \text{ is smoking} \end{cases}$
- Y_i = weight of baby i

Critical assumption on the "error" term:

$$E(\varepsilon_i | X_i) = 0$$

One can show that if $Cov(\varepsilon_i, X_i) \neq 0$

then $E(\varepsilon_i | X_i) \neq 0$

Why is this problematic?

Answer: We can only identify β_2 if $E(\varepsilon_i | X_i) = 0$

since: $Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i$

Regression analysis estimates $E(Y_i | X_i)$

So $E(Y_i | X_i) =$

$$E(\beta_1 + \beta_2 X_i + \varepsilon_i | X_i) =$$

$$E(\beta_1 + \beta_2 X_i | X_i) + \underbrace{E(\varepsilon | X_i)}_{= 0 \text{ if our assumption is correct.}} =$$

deterministic → $\beta_1 + \beta_2 X_i$ ("=" $\underbrace{E(X_i | X_i)}_{= X_i}$)

Instrumental variables can fix this problem.

Problem: $Cov(\varepsilon_i, X_i) \neq 0$

$\Rightarrow E(\varepsilon_i | X_i) \neq 0$

$\Rightarrow$ We cannot identify β_2

$\Rightarrow$ we cannot estimate β_2

2. Robust Standard Errors

Classic regression model:

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i$$

$$(Y_i - \beta_1 - \beta_2 X_i = \varepsilon_i)$$

Assumptions on ε_i:

- $E(\varepsilon_i | X_i) = 0$

"Mean of ε_i is zero no matter what value X_i takes."

- Classic assumption on ε_i:

$$Var(\varepsilon_i | X_i) = \sigma^2 \text{ for all } i = 1, \dots, n$$

(σ^2: same for all)

Homogeneous variance of ε_i

- More realistic: Heterogeneous variance of ε_i: $Var(\varepsilon_i | X_i) = \sigma_i^2$

3. Non-Parametric Density Estimation

We consider data to be generated from a "random sample":

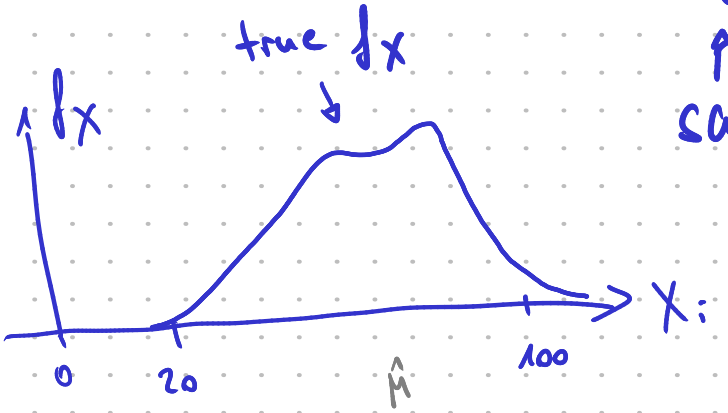
$$\{X_1, X_2, \dots, X_n\}$$

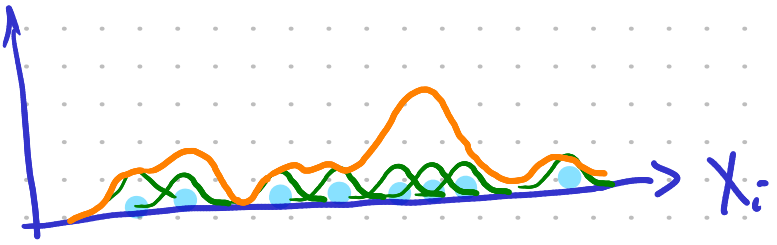
E.g. X_i = working-time / week of CEO i

$i = 1, \dots, n = 200$

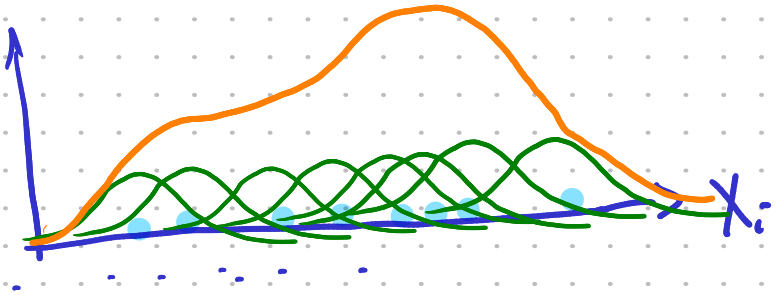
Assumption: $X_i \sim f_X$ ← density of X_i

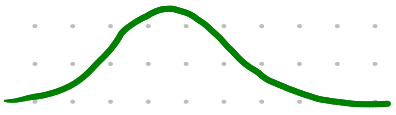
↑ same for all $i = 1, \dots, n = 200$





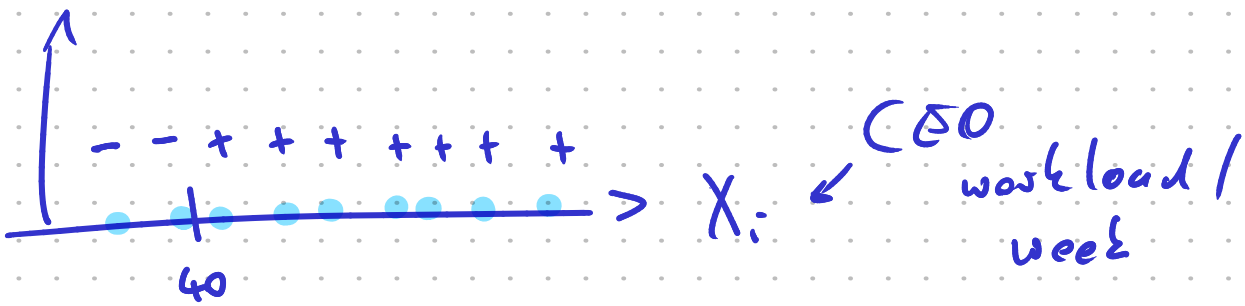
too 'small' bump





$$\hat{f}_X(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)$$

Rank-based testing



H_0 : Median of X_i is 40

H_1 : Median of X_i is not 40

Homework:

- Form groups (2 or 3 groups)
- Find topic(s)
- Setup Python on your <u>own</u> PC.

Next meeting: Monday Nov 16

9:30

https://www.dliebl.com/Script-Coding-Intros/